power

File: power.kicad_sch

pi_link

File: pi_link.kicad_sch

radio

File: radio.kicad_sch

ui

File: ui.kicad_sch

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MTP F26 Team C (UPC)

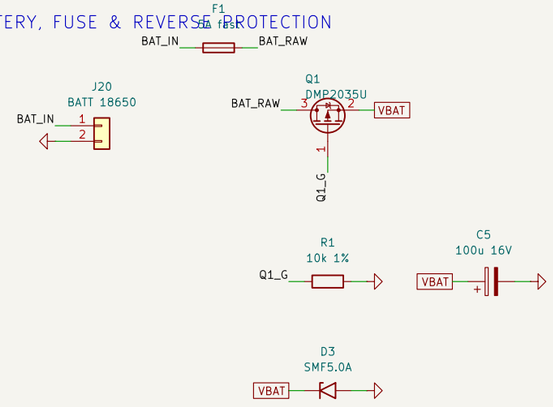
Sheet: /
File: teamc-carrier-lab.kicad_sch

Title: Team C carrier rev B - UPC lab, hand-soldered modules (DRAFT)

Size: A4	Date: 2026-10-19	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/5

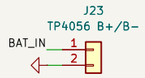
POWER - one protected 18650 → 5 V for the Pi 3 A+ (Pololu U3V70F5) and 3.3 V for the radio (AP2112K)

1 BATTERY, FUSE & REVERSE PROTECTION



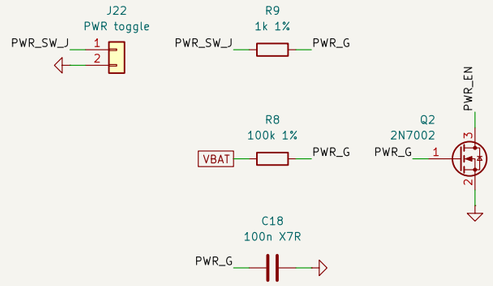
Protected 18650 (PCM inside, trip >= 3 A) in a holder on the box floor, JST-VH lead to J20.
Q1: body diode conducts for a correct cell, then VGS = −VBAT turns it on; a reversed cell is blocked.
DMP2035U <= 45 mΩ at −2.5 V (AO3401A ~85 mΩ: -0.25 V at 3 A). C5: bulk at the boost input (Pololu asks for >= 33 uF when the cell is on a lead). Charger (J23): rev B on BAT_IN, beside J20 (the cell lead).

2 CHARGER LEAD (TP4056 USB-C module at the box wall)



TP4056 USB-C module with protection (HW-373), off-board where the USB-C is reachable:
its B+/B− to this lead (1 A charge; XH is rated 3 A). OUT+/OUT− unused.
No power path: charge with the box switched OFF. Check its CC resistors with a C-to-C cable.

3 POWER TOGGLE → BOOST EN (closed = ON)

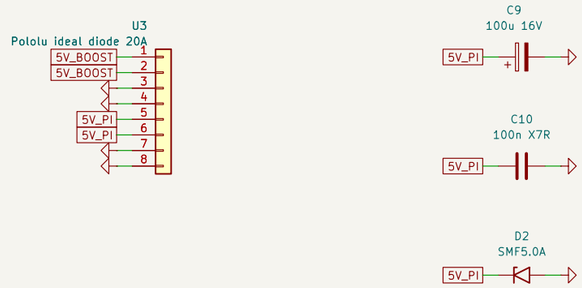


Toggle closed: PWR_G = 0 → Q2 off → the module's own 100 k pull-up holds EN high → boost ON.
Toggle open (or a broken lead): R8 → Q2 on → EN low → boost OFF (true shutdown, output disconnected).
~40 µA either way. R9/C18: filter + ESD for the panel lead.

4 5 V BOOST MODULE (Pololu U3V70F5)

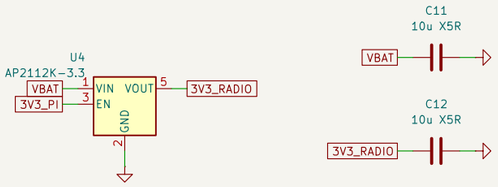
~2.9 A out at 3.0 V in (Pololu graph), 5.0 V ±4 %, reverse-voltage protected, UVLO 2.4/2.8 V, Iq < 1 mA.
ENABLE: 100 k pull-up to VIN on the module, < 0.4 V = true shutdown (load disconnected).
High-current loop: J20 → F1 → Q1 → U2 VIN, U2 GND → J20 GND: short and wide (1.5 mm), star point at J20.
Keep U2 at the opposite end of the board from J30 (radio).

5 BACK-POWER IDEAL DIODE (Pololu #5388) & 5V_PI AT THE IDC HEADER



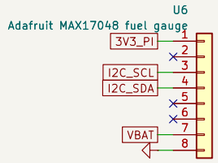
< 3 mΩ: the Pi sees 5.0 V − ribbon (~0.03 V at 1 A) > 4.63 V throttle. A Schottky here would not do.
LM74700 inside: needs ~3.9 V to start (runs down to 3.2 V) − fine here on the 5 V side, never on the cell.
C9/C10/D2 at J1 pins 2/4.

6 RADIO SUPPLY: AP2112K-3.3 (EN = the Pi's 3V3)



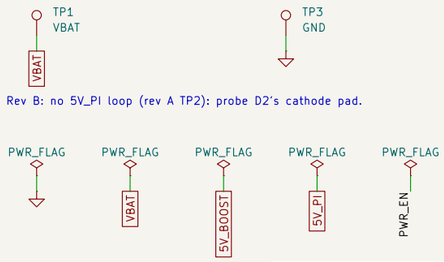
~0.1 V dropout at 150 mA (600 mA part). EN from 3V3_PI: the radio is off whenever the Pi is off
(else the SP4 in RX drains ~26 mA from the cell with the box 'off'). The breakout adds 110 uF at the module.
C12 10 uF (rev A 1 uF + 10 uF C71 at the socket): the socket is 6 mm away on the lab board; AP2112 needs >= 1 uF.

7 FUEL GAUGE (Adafruit MAX17048 #5580, I2C 0x36)



The chip runs from the cell (SJ2 bridged VDD-Bat as shipped); header VIN only feeds its 10 k pull-ups on
SCL/SDA/INT and the LED (cut SJ1 to save ~0.1 mA): VIN = 3V3_PI (rev B: from J1 pin 1). Cell sense through the JP2 Bat/GND
pair (2-pin header, Kelvin track from Q1's source), not a JST-PH lead. QSTART unused (10 k to GND on
the module). INT (alert) not routed in rev B: the software polls SOC over I2C and on the single
copper layer the line would need a long wire link; a flying wire INT -> J1 pin 22 (GPIO 25) adds it.
The module's INT pull-up stays on its board. Rev B: the header GND pin is not connected − the module's
ground is the JP2 BAT_GND pin (same copper on the Adafruit board), tied to the carrier ground.
No level shifting needed at 3.3 V. Software: never set MODE.EnSleep.

8 TEST POINTS



Rev B: no 5V_PI loop (rev A TP2): probe D2's cathode pad.

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DRAFT - for ET review
MTP F26 Team C (UPC)

Sheet: /power/
File: power.kicad_sch

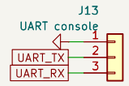
Title: Power: cell, boost/ideal-diode/fuel-gauge modules, radio LDO (rev B

Size: A3	Date: 2026-10-19	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/5

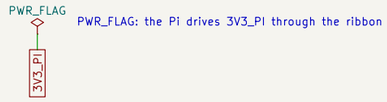
PI LINK - Raspberry Pi 3 Model A+ by a 40-way ribbon

DRAFT - for ET review. Rev B is not a HAT: on the lab's unplated holes a socket under a HAT would have to be soldered on the top side while the modules solder on the bottom. J1 is a shrouded 2x20 box header soldered on the copper side like every other THT part; odd pins leave to one side, even pins to the other.
No ID EEPROM: pins 27/28 stay unconnected. The Pi's WiFi/BT chip is disabled or desoldered (Atenea Q22).
5V_PI comes from the power sheet's ideal diode. Rule as in rev A: never plug the Pi's micro-USB while the carrier is connected and switched on (the ideal diode blocks back-feed into the boost, but two 5 V sources would still share the Pi). 3V3_PI = the Pi's own 3.3 V, rev B from pin 1 (next to the fuel gauge; pin 17 is unused): the fuel gauge's VIN and the radio LDO's EN only. Rev B ground: the carrier ties the Pi's GND on pins 6, 9 and 14; pin 25 is the LEDs' return (LED_RTN); pins 20/30/34/39 sit in pockets of the one-layer escape tracks, unused.

UART CONSOLE



1 GND, 2 TX (Pi GPIO 14 out), 3 RX (GPIO 15 in). 3.3 V levels only.



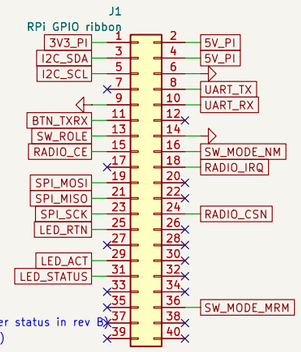
PWR_FLAG: the Pi drives 3V3_PI through the ribbon

MOUNTING (M3, unplated, keep copper away)



Header pin / BCM function (pinout.xyz)

1	3V3 (rev B: pin 1 feeds 3V3_PI)
3	GPIO 2 SDA1
5	GPIO 3 SCL1
7	GPIO 4
9	GND
11	GPIO 17
13	GPIO 27
15	GPIO 22
17	3V3 (rev B: unused, pin 1 instead)
19	GPIO 10 MOSI0
21	GPIO 9 MISO0
23	GPIO 11 SCLK0
25	GND (rev B: the LEDs' return only)
27	ID_SD (GPIO 0): keep NC
29	GPIO 5
31	GPIO 6
33	GPIO 13
35	GPIO 19
37	GPIO 26 (rev A CHG_STAT; no charger status in rev B)
39	GND (rev B: not used on the carrier)



Header pin / BCM function

2	5V
4	5V
6	GND
8	GPIO 14 TXD0
10	GPIO 15 RXD0
12	GPIO 18
14	GND
16	GPIO 23
18	GPIO 24
20	GND (rev B: not used on the carrier)
22	GPIO 25 (rev A FG_ALRT; not routed in rev B)
24	GPIO 8 CE0
26	GPIO 7 CE1
28	ID_SC (GPIO 1): keep NC
30	GND (rev B: not used on the carrier)
32	GPIO 12
34	GND (rev B: not used on the carrier)
36	GPIO 16
38	GPIO 20
40	GPIO 21

WIRE LINKS (rev B, single copper layer): bare 0.6 mm tinned copper wire on the component side between two unplated 1.0 mm holes, soldered on the copper side; the board's crossings.
Positions and lengths: hw/teamc-carrier-lab/links.json.



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Sheet: /pi_link/
File: pi_link.kicad_sch

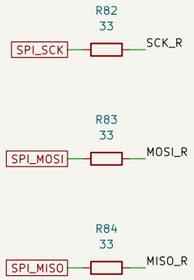
Title: Pi link: 2x20 IDC box header + ribbon, UART, holes (rev B lab, DRAFT)

Size: A3	Date: 2026-10-19	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/5

RADIO - socket for the lab-made E01-ML01SP4 breakout

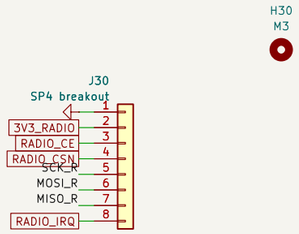
DRAFT - for ET review.
VCC 2.0-3.6 V, ABS. MAX 3.6 V: only 3V3_RADIO from the AP2112K on the power sheet (enabled by the Pi's 3V3, so the radio is off whenever the Pi is off). Never the cell, never the Pi's 3V3 pin.
The breakout has the module's 100 uF + 10 uF + 100 nF. Rev B: no socket-end cap (rev A C71) and no test loop (rev A TP10): C12 (10 uF) at the LDO is 6 mm from the socket, and 3V3_RADIO is probed on C12's pad.
Place J30 so the breakout's IPEX end faces the board edge, far from the boost module; M3 standoff at the breakout's H1 (H30 here, ~11 mm standoff = socket + header height). Antenna: TX2400-FPC-2509 flex on a box wall on the U.FL (glue the plug, drop test).

SPI SOURCE TERMINATION (33 Ω)



R82 SCK, R83 MOSI: at the IDC header J1 (drivers = Pi, through the ribbon).
R84 MISO: at J30 (driver = SP4). The ribbon adds ~10 cm: keep these (0 Ω if the bench shows no need).

BREAKOUT SOCKET



PULL-UP TO 3V3_RADIO



R80: CSN high = radio deselected while the Pi boots.
Rev B: to 3V3_RADIO (rev A: 3V3_PI), so it sits at the socket without a 3V3_PI track there on the single copper layer. 3V3_RADIO comes up with 3V3_PI (the LDO's EN is 3V3_PI): CSN is still high from the first milliseconds of the Pi's boot, and both rails are off together.
No IRQ pull-up in rev B (rev A R81): the Linux HAL requests the Pi's pull-up on GPIO 24 (IRQ is open drain, falling edge only: radio/fw/linux/nrf24_hal_linux.c), and on one copper layer 3V3 does not reach pin 8.

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Sheet: /radio/
File: radio.kicad_sch

Title: Radio: SP4 breakout socket (rev B lab, DRAFT - for ET review)

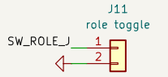
Size: A3	Date: 2026-10-19	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/5

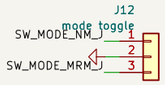
UI - panel controls on JST-XH leads, bench button, LEDs

DRAFT - for ET review. Software: proto/port/linux/ui.c (inputs active low with the Pi's pull-ups; LEDs active high).
Mode switch (J12, ON-OFF-ON): GPIO 23 low = NM, else GPIO 16 low = MRM, else (centre) SRI. Role (J11): low = TX.
Pull-ups: the Pi's own (ui.c); rev A's external 10 kΩ are gone in rev B (one copper layer). 1 kΩ series: GPIO 16 may be driven by legacy firmware at boot (HAT design guide); 1 kΩ limits that to 3.3 mA into a closed switch. A TVS per lead right at the connectors.
Panel holes: Ø7.5 mm for the PS1024 button, Ø6.5 mm for the 100-series toggles.
Glue the XH housings to the board and the leads to the box (drop test).

CONNECTORS (panel parts off-board)







J10: panel button E-Switch PS1024ABLK (momentary).
Bench: plug the panel button into J10 (or bridge its two pins); no on-board button.
J11: role toggle E-Switch 100SP1T1B1M1QEH, closed = TX;
  may stay unfitted until Atenea Q8 (open = RX).
J12: mode toggle E-Switch 100SP3T1B1M1QEH ON-OFF-ON:
  1 NM, 2 common (GND), 3 MRM; centre = SRI.

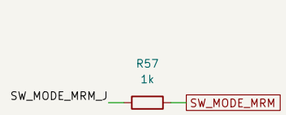
NO EXTERNAL PULL-UPS IN REV B (R50-R53 of REV A) 1 kΩ

The inputs use the Pi's pull-ups, which ui.c requests (GPIO_V2_LINE_FLAG_BIAS_PULL_UP) before it reads them. Until then (boot) the BCM2837 default pull-down makes an open toggle read 'active' - nothing reads them then. On one copper layer a 3V3 bus to four pull-ups at the leads cuts the ground pour into islands.

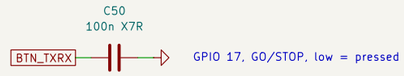


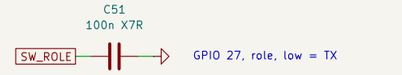


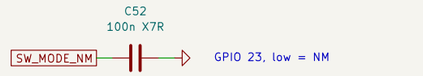


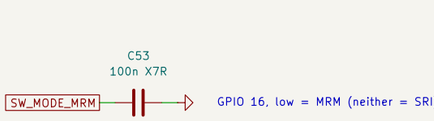


100 nF AT THE GPIO









ESD (one TVS per lead, at the connectors)



Rev B: four SMF5.0A (5 V standoff, 200 W, pin 1 = cathode to the line) - the same already used for D2/D3, instead of rev A's SP0504 4-line array: on one copper layer a single SOT-23-5 cannot reach four parallel lead lines without a crossing. The TVS capacitance (~1 nF) sits before the 1 kΩ / 100 nF filter: harmless.

LEDs (active high, ≈ 4 mA from 3.3 V)



D10 activity (GPIO 5), D11 status (GPIO 6).
(3.3 - ~2.0 V) / 330 Ω ≈ 4 mA, well under the
Pi's 16 mA per pin. No InGaN (blue/white: Vf too high).
Rev B: the cathodes return on LED_RTN = J1 pin 25 (a Pi GND pin) - on one copper layer the LEDs sit in a pocket of J1's escape tracks, away from the carrier's pour.

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MTP F26 Team C (UPC)

Sheet: /ui/		
File: ui.kicad_sch		
Title: UI: panel button, toggles, LEDs (DRAFT - for ET review)		
Size: A3	Date: 2026-10-19	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/5